

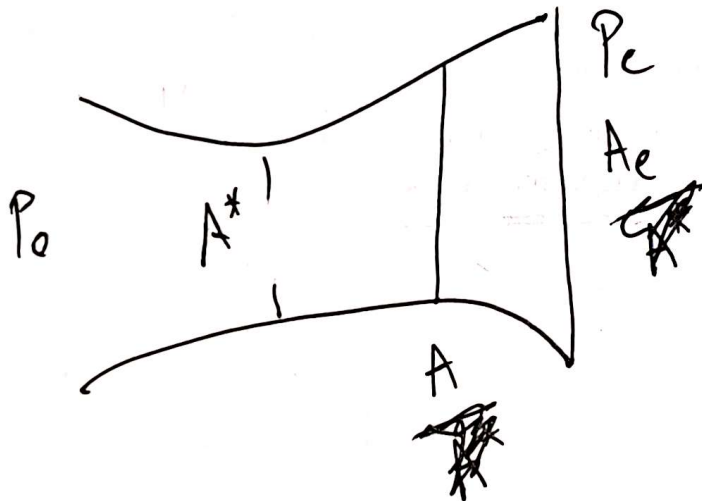
Use 3.15, put $\frac{A_e}{A^*}$ get M at A_e (sup M_e)

(2)

Use $\frac{P_e}{P_0} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}}$ get $\frac{P_e}{P_0}$

that gives you M_e , so if you want

M_e , set $\frac{P_e}{P_0}$ from (2)



$$f\left(\frac{A_e}{A^*}\right) = M_e \quad (3.15)$$

$$f\left(\frac{P_e}{P_0}\right) = M_e \quad (2)$$

Both should be satisfied

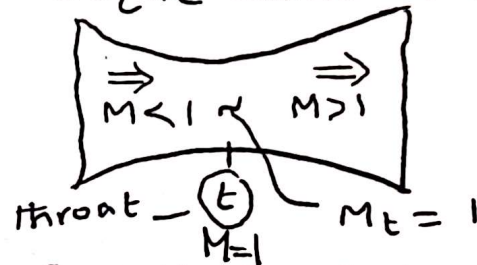
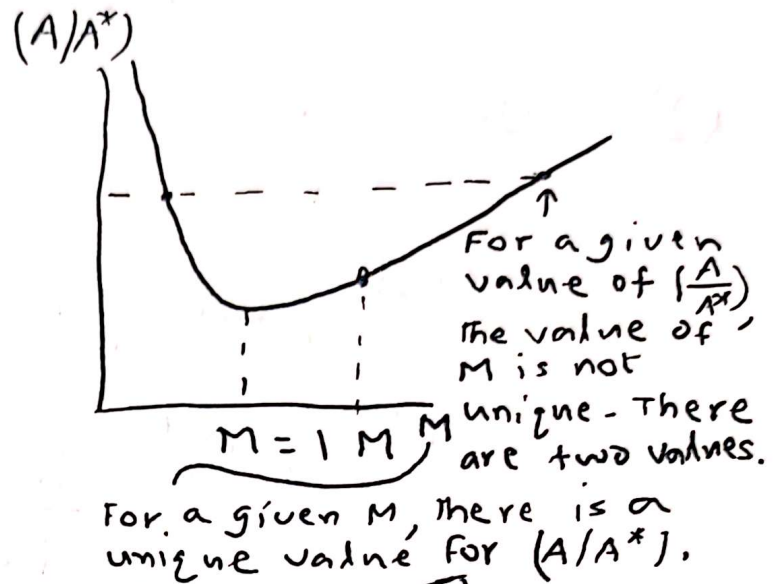
$\Rightarrow$ Perfect Expansion

$$P_1 + \frac{\rho_1 u_1^2}{2} = P_2 + \frac{\rho_2 u_2^2}{2}$$

$$T = \dot{m} u_e + \underline{\underline{(p_e - p_a) A_e}}$$

Comments:

- To accelerate the flow from $M < 1$ to $M > 1$, a c-d passage is required.
- A converging passage can accelerate flow from $M < 1$ to a maximum $M = 1$.
- The Mach number at the throat (the minimum area location) of a c-d nozzle, in which the flow accelerates from $M < 1$ to $M > 1$, will be 1.
- A nozzle operating with $M = 1$ at its throat is said to be choked.



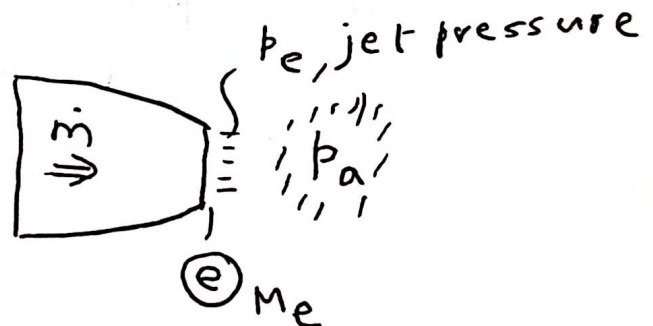
Important observation about subsonic jets-

The pressure of a subsonic jet, issuing out of a confinement into unconfined space, is equal to the pressure in the unconfined space.

Convergent nozzle

If $M_e < 1$, $p_e = p_a$.

If $M_e = 1$, $p_e > p_a$.



c-d (de Laval) nozzle

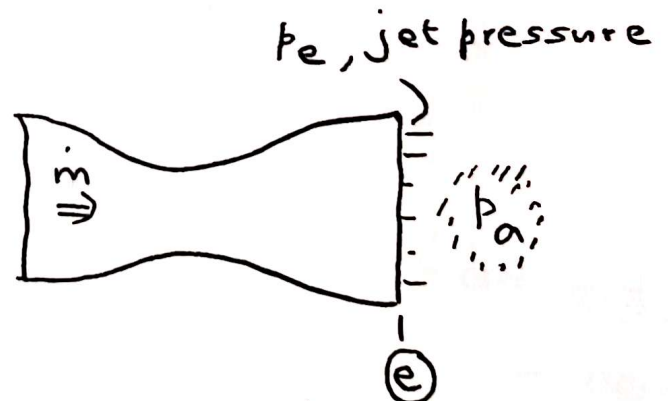
If $M_e < 1$, $p_e = p_a$.

If $M_e > 1$, p_e may not be equal to p_a .

$p_e > p_a$ (under expanded jet)

$p_e = p_a$ (fully expanded jet)

$p_e < p_a$ (overexpanded jet)



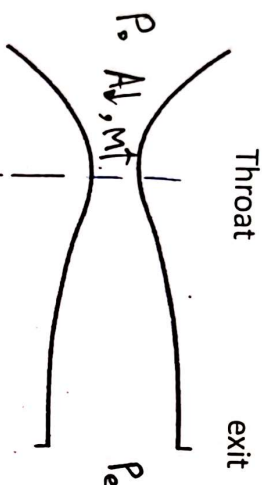
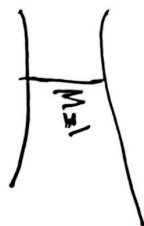
p_a is pressure in ambient (unconfined) space.



$$T = m u_e + (p_e - p_a) A_e$$

$$(A/A^*)_{1,1}$$

$$p_1 + \gamma u_1^2 = p_2 + \gamma u_2^2 \uparrow$$

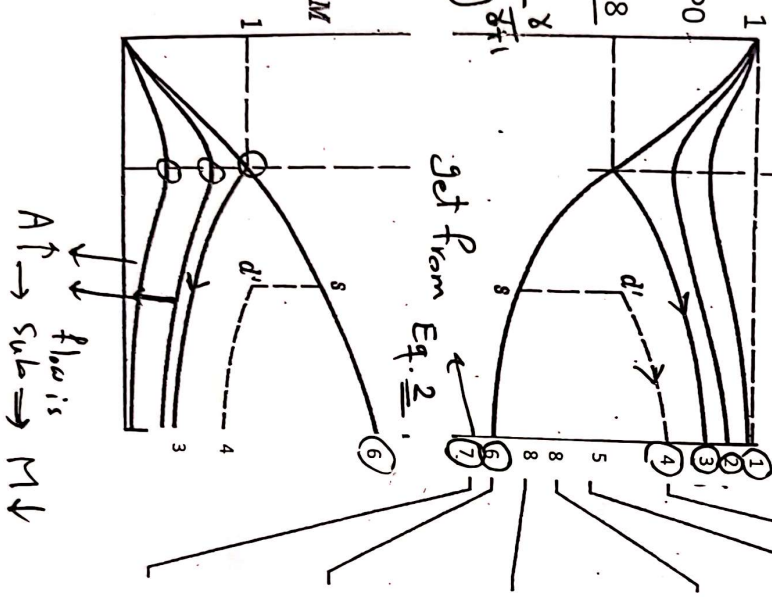


$$\frac{P_t}{P_0} \approx 0.528$$

$$\frac{P_t}{P_0} = \left(1 + \frac{\gamma-1}{2} M_t^2\right)^{-\frac{\gamma}{\gamma-1}}$$

$$M_t = 1 \Rightarrow$$

$$\frac{P_t}{P_0} = \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma}{\gamma-1}} = 0.528$$



Get from Eq. 2.1



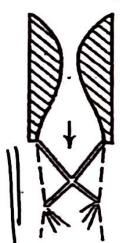
Shock in nozzle



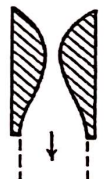
Shock at exit



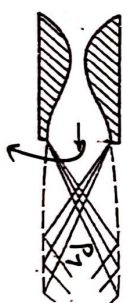
Overexpanded, oblique shock



Overexpanded, oblique shock



Perfect expansion



Underexpanded, expansion fan